LA-9: Practice Data Analysis (15 points)

Learning Outcomes

In this assignment, you will practice the data analysis skills that you have learned this semester:

- Calculating descriptive statistics for groups with `descr()` (LA-4 to LA-8).
- Selecting, conducting, and interpreting a statistical test, choosing among all four tests: chi-squared test, *t*-test, ANOVA, and Pearson's correlation (LA-7 and LA-8).
- Reporting test results in APA format (LA-7 and LA-8).

> **Tip**
>
> Read all the instructions carefully before starting the assignment. Set up your R script as you did in LA-4 to LA-8 (name, uNID, date, assignment number).

> **Important**
>
> **How to report your answers.** Write each answer as a comment in your R script, labeled with its question number (e.g., `# 3a: ...`). Report test results in APA format, as shown in the section "Conducting a Statistical Test in R" at the end of this document. Round means and standard deviations to two decimal places and include units (e.g., $7.50).

> **Note**
>
> **Key ideas (from LA-7 and LA-8):**
>
> - The **independent variable (IV)** is the variable we think *explains* or *predicts* differences. The **dependent variable (DV)** is the outcome we are measuring.
> - A **categorical** variable sorts cases into groups. A variable coded with numbers like 0 and 1 can still be categorical if the numbers stand for groups (like `majority_life_impacted` in LA-8). A **continuous** variable is a number that can take many values.
> - Use the flowchart in the section "Selecting a Statistical Test" at the end of this document to choose a test. Start at the top with the DV.
> - In this course, if $p < .05$, we conclude that there **is** a statistically significant difference (or relationship). If $p \geq .05$, we conclude that we **do not** have evidence of a difference (or relationship).

Instructions

1) Load the packages below. Then download the CPS dataset (`CPS.csv`) and its codebook (`CPS-codebook.csv`) from Canvas, upload the dataset to your Posit Cloud project (as you did in LA-4 to LA-8), and read it into R. Use the codebook to familiarize yourself with the variables in the data.

- `tidyverse`
- `summarytools`
- `rstatix` (if R says it can't find the package, run `install.packages("rstatix")` first)

> **Note**
>
> **About this dataset.** Data for this assignment come from the 1985 Current Population Survey (CPS). The CPS is used to supplement census information between census years. These data are a random sample of 534 workers from the survey, with information on their wages and other characteristics, including sex, years of education, years of work experience, job sector, region of residence, and union membership.
> Unlike the COVID data in LA-7 and LA-8, each row (case) in this dataset is **one person**.

2) Obtain descriptive statistics for wages earned per hour (`wage`) for females and males (`sex`).

   a) What is the mean and standard deviation of hourly wages for females?
   b) What is the mean and standard deviation of hourly wages for males?

> **Tip**
>
> To get descriptive statistics for each group, use `group_by()` before `descr()`, as in LA-7 and LA-8, or use `filter()` and `descr()` for each group.

3) Conduct a statistical test to determine whether there is a significant difference in the average hourly wages of males and females.

   a) State your hypothesis.
   b) Select and conduct a statistical test to address your hypothesis.
   c) Report the test statistic and p-value in APA format.
   d) Does the statistical test support your hypothesis? If there is a significant difference, describe it (e.g., do males or females earn higher wages, on average?).

> **Tip**
>
> As in LA-7 and LA-8, `t_test()` subtracts the mean of the second group from the mean of the first group (groups are in alphabetical order, so `F` comes before `M`). A **negative** t means the first group's mean is **lower**. Use your answers to Question 2 to see which group is higher. You can (and probably should) confirm this by checking the mean values in each group.

4) Select and conduct the appropriate statistical test to determine whether there is a significant linear relationship between wages earned per hour (`wage`) and number of years of work experience (`exper`).

   a) State your hypothesis.
   b) Select and conduct a statistical test to address your hypothesis.
   c) Report the test statistic and p-value in APA format.
   d) Does the statistical test support your hypothesis? If there is a significant relationship, describe it (e.g., is it positive or negative? Is it strong or weak?).

> **Tip**
>
> When you are looking for a relationship between two continuous variables, it does not matter which variable you list first in `cor_test()`.

5) Is job sector (`sector`) related to whether workers are satisfied with their current job (`satisfaction`)?

   a) State your hypothesis.

b) Select and conduct a statistical test to address your hypothesis.
c) Report the test statistic and p-value in APA format.
d) Does the statistical test support your hypothesis? Explain.

> **Tip**
>
> `satisfaction` is coded 0 and 1, but the numbers stand for groups (1 = satisfied with current job; 0 = not satisfied). `sector` has eight groups; run `cps |> freq(sector)` to see them.
> The chi-squared command uses `cps$sector`, which means "the `sector` column of the `cps` dataframe," as in LA-8.

6) Is there a significant difference in average hourly wages (`wage`) across job sectors (`sector`)?

   a) State your hypothesis.
   b) Select and conduct a statistical test to address your hypothesis.
   c) Report the test statistic and p-value in APA format.
   d) Does the statistical test support your hypothesis? Explain. Then use `group_by()` and `descr()` to find which job sector has the highest average hourly wage and which has the lowest.

Submission

Submit your R script (named `LA-#_FirstName-LastName.R`) to Canvas.

Your R script should:

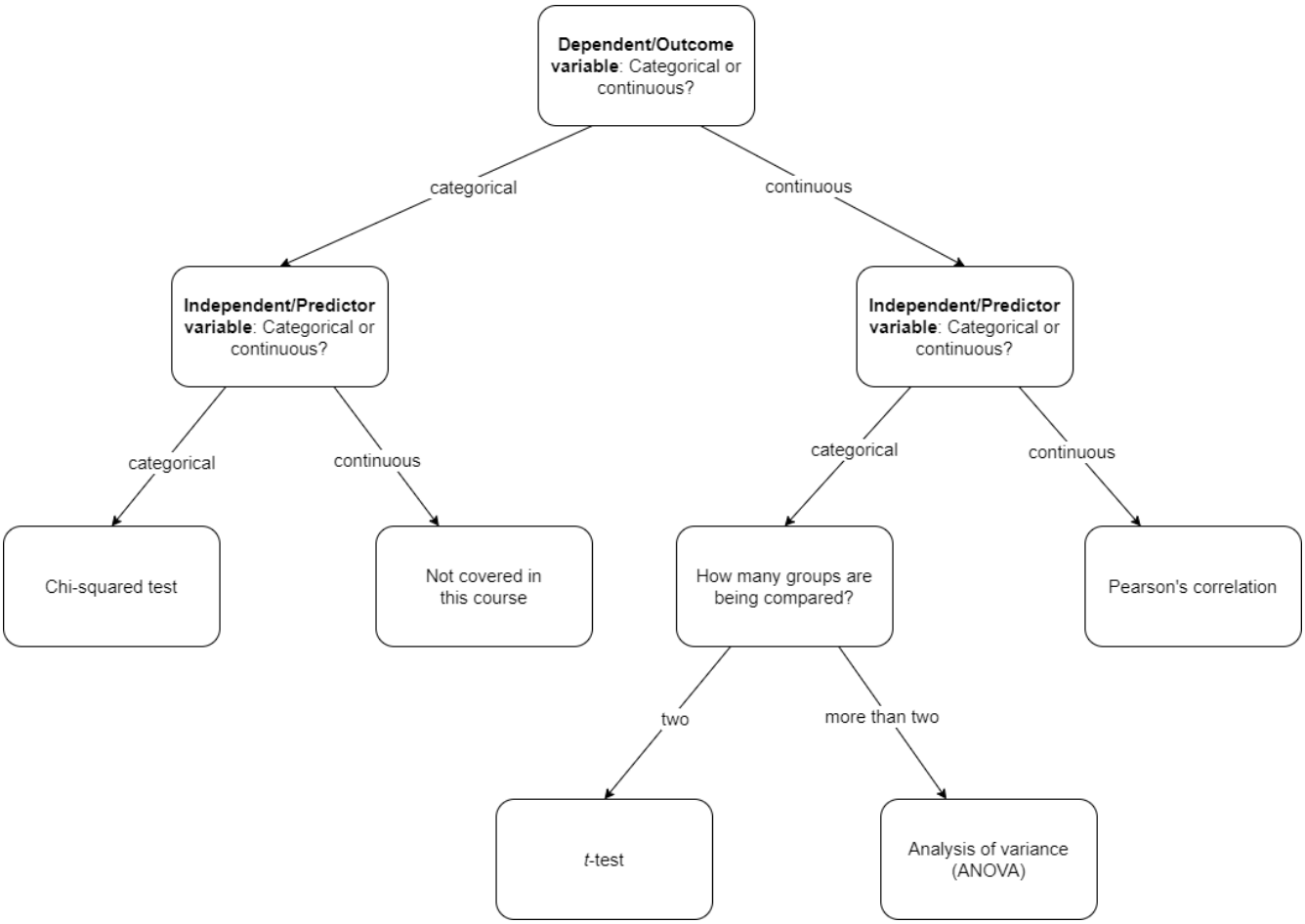
1) Include commands and functions that are necessary to address all the questions in the assignment.
2) Contain comments that answer the questions in the assignment.
3) Run in its entirety without errors.

To ensure that your R script runs without errors, you should:

- Save your script.
- Navigate back to Your Workspace on Posit Cloud.
- Reopen your project.
- Run the entire script line-by-line without editing it to ensure there are no errors.

> **Important**
>
> These standards apply to all submissions in this course that require R scripts. You should follow these instructions for preparation, naming, and saving of your R script for all of your individual lab assignments.

Selecting a Statistical Test



Conducting a Statistical Test in R

The example output below is **made up** for illustration, so your numbers will be different. Use these examples to see what the output looks like and how to report it.

Chi-squared test

A chi-squared test is used when we want to compare two categorical variables. To do so in R, you can use the following command (`df`, `dep_var`, and `ind_var` have to be replaced with the names of your dataframe, dependent variable, and independent variable, respectively). The output that results from running this command in R is shown below as comments.

```
chisq_test(x = df$ind_var, y = df$dep_var)

# A tibble: 1 × 6
#       n statistic         p    df method          p.signif
# * <int>     <dbl>     <dbl> <int> <chr>           <chr>
# 1   200      6.84    0.0327     2 Chi-square test *
```

To report the results of your chi-squared test in APA format, you would use the following format, replacing the

text in bold font with the results from your Console:

$\chi^2(\mathbf{df}, N = \mathbf{n}) = \mathbf{statistic}, p = \mathbf{p}$

From the results of the above chi-squared test, I would report results like this: $\chi^2(df = 2, N = 200) = 6.84, p = .033$. If the p-value is less than .001, you do not need to report the exact value. If it is greater than .001, then report the exact value rounded to two or three decimal places.

t-test

A t-test is used to compare the mean values of a continuous variable between two groups. To do so in R, you can use the following command (`df`, `dep_var`, and `ind_var` have to be replaced with the names of your dataframe, dependent variable, and independent variable, respectively). The output that results from running this command in R is shown below as comments.

```
df |>
        t_test(dep_var ~ ind_var)

# A tibble: 1 × 10
#   .y.        group1 group2    n1    n2 statistic    df        p    p.adj p.adj.signif
# * <chr>      <chr>  <chr>  <int> <int>     <dbl> <dbl>    <dbl>    <dbl> <chr>
# 1 dep_var ind_var1  ind_var2  48    52     -4.12  97.6 7.94e-05 7.94e-05 ****
```

To report the results of your t-test in APA format, you would use the following format, replacing the text in bold font with the results from your Console:

$t(\mathbf{df}) = \mathbf{statistic}, p = \mathbf{p}$

Using the results above, I would report the results of a t-test like this: $t(df = 98) = -4.12, p < .001$. Round `df` to a whole number if it has decimals.

Analysis of variance (ANOVA)

ANOVA is used when we want to compare the mean values of continuous variable between more than two groups. To do so in R, you can use the following command (`df`, `dep_var`, and `ind_var` have to be replaced with the names of your dataframe, dependent variable, and independent variable, respectively). The output that results from running this command in R is shown below as comments.

```
df |>
        anova_test(dep_var ~ ind_var)

# ANOVA Table (type II tests)
#
#   Effect  DFn DFd     F        p p<.05   ges
# 1 ind_var   2 450 8.936 0.000156     * 0.038
```

To report the results in APA format:

F (`DFn`, `DFd`) = `F-value`, p = `p-value`

Using the results above, I would report: $F(2, 450) = 8.94, p < .001$.

Correlation

We use Pearson's correlation to examine a linear relationship between two continuous variables. To run a correlation in R, you can use the following command (`df`, `dep_var`, and `ind_var` have to be replaced with the names of your

dataframe, dependent variable, and independent variable, respectively). The output that results from running this command in R is shown below as comments.

```
df |>
        cor_test(dep_var, ind_var)

# # A tibble: 1 × 8
#   var1          var2                cor statistic        p conf.low conf.high method
#   <chr>         <chr>             <dbl>     <dbl>    <dbl>    <dbl>     <dbl> <chr>
# 1 dep_var       ind_var            0.31      4.59 0.00000793   0.178     0.431 Pearson
```

To report the results in APA format:

> Pearson's $r =$ `cor`, $p =$ `p-value` (or $p < .001$ if the p-value from your test is very small, i.e., less than .001)

Using the results above, I would report: Pearson's $r = 0.31$, $p < .001$.